DAXALGO RESEARCH // VERIFIED INPUT BUNDLE

FDT-N volume-footprint transformer

Self-contained research paper and frozen multi-seed evaluation | BTCUSDT spot

EVIDENCE BOUNDARY

FORECAST EVALUATION - NOT A TRADING SYSTEM

The primary run is selected by validation loss only. Every supplied seed remains in the held-out scorecard; no test metric participates in run selection.

3	17	0.11263	0.07039
DECLARED SEEDS	PRIMARY SEED	VALIDATION LOSS	HELD-OUT TEST LOSS
all reported	validation-only choice	selection criterion	reported after selection

Frozen corpus: 2024-01-01 through 2024-01-14 UTC

Results SHA-256: 746403a05790753943b8c22f615f097c8dc397e413be49f35ea553b82e0cbd0a

ABSTRACT

Abstract and evidence grammar

Architecture facts, hypotheses, and measured results remain visibly separate.

OBSERVED

The market object

A sealed footprint bar retains executed buy and sell volume at every traded price row, plus POCs, centroids, range, value area, volume, delta, CVD, imbalance, quality, and UTC time. The frozen corpus identity is measured independently of forecast skill.

IMPLEMENTED

The forecaster

FDT-N embeds every accepted row, pools each bar with four learned queries, fuses the row summary with 23 aggregates, processes 64 sealed bars, and directly decodes eight future horizons. Each horizon emits q10/q50/q90 for seven target families.

DESIGN HYPOTHESIS

Why test this hierarchy

Full auction shape may preserve information discarded by aggregates; learned query pooling may summarize variable row sets efficiently; direct horizon queries may avoid autoregressive forecast-as-input error. None of these motivations is a result.

EMPIRICAL RESULT

What this edition measures

The report binds 3 declared seed results to one frozen dataset, selects the primary only by validation loss, publishes every held-out seed, and binds the selected checkpoint to a passed ONNX Runtime parity artifact. It remains a short, single-instrument development study.

Research questions

- RQ1 - Do exact row distributions improve proper held-out scores over aggregate-only inputs?
- RQ2 - Do four learned queries improve on simple pooling or convolutional structure?
- RQ3 - Do parallel horizon queries improve on separate heads and naive causal baselines?
- RQ4 - Are q10/q50/q90 calibrated overall and by horizon, target, and regime?
- RQ5 - Does any gain survive other symbols, venues, regimes, and runtime constraints?

SAFETY AND CLAIM BOUNDARY

FDT-N is a research forecaster and chart-visualization candidate. It creates no order, position, sizing instruction, or execution signal. Forecast accuracy is not economic value; fees, latency, fills, impact, and a frozen decision policy require a separate study.

BEGINNER PRIMER

Volume footprints from first principles

Executed-flow observations describe an auction; they are not directional conclusions.

OBSERVED

One price row

At price row j inside sealed bar t, p is snapped price, v+ is buy-initiated ask-lifting volume, and v- is sell-initiated bid-hitting volume. Bid/ask imbalance and zero-side flags are explicit data fields, not inferred tiny volumes.

ROW IDENTITIES

r[t,j] = (p[t,j], v+[t,j], v-[t,j], imbalance flags, zero flags)

total row volume: v[t,j] = v+[t,j] + v-[t,j]

row delta: Delta[t,j] = v+[t,j] - v-[t,j]

INTERPRETATION

Delta is descriptive

Positive delta says aggressive buyers executed more volume at that row or bar. It is not automatically bullish: price response, passive absorption, regime, and later flow are separate objects.

OBSERVED

From rows to a sealed bar

Bar buy and sell volumes sum the corresponding row volumes. Total POC is the price row with maximum total volume; buy and sell POCs use side volume. A centroid is a weighted mean, not an argmax. Value area is a deterministic contiguous region grown around POC.

BAR IDENTITIES

V+ = sum_j v+[j] ; V- = sum_j v-[j] ; V = V+ + V-

Delta = V+ - V- ; POC = p[argmax_j(v[j])]

centroid = sum_j(p[j] * v[j]) / V

CVD[t] = CVD[t-1] + Delta[t]

DATA CONTRACT

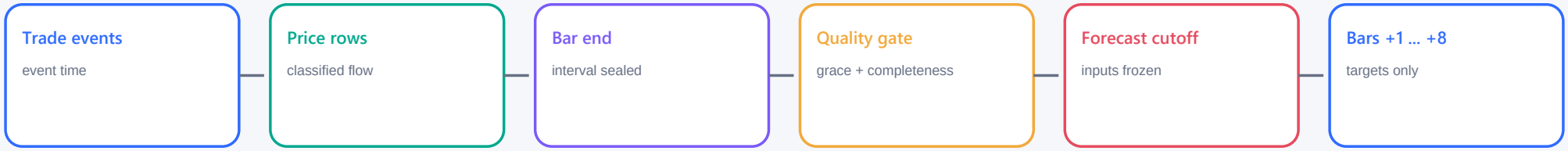
Policies are part of the coordinate

Tie handling, price snapping, aggressor classification, value-area expansion, CVD reset, bar interval, row size, and feed-quality decisions must be deterministic and versioned. Changing any of them changes the meaning of the tensor.

PROBLEM DEFINITION

Causal cutoff and forecasting task

Only information available before the cutoff may influence inputs, normalization, or selection.



DIRECT MULTI-HORIZON MAP

Y_hat[t+1:t+8] = F_theta(B[t-63], ..., B[t])

output shape = [batch, 8 horizons, 7 targets, 3 quantiles]

no realized target and no earlier forecast is fed back into a later horizon

Family	Targets	Coordinate
Price	total POC, low, high, buy POC, sell POC	cutoff-POC offsets
Activity	log total volume	context-median relative
Flow	delta fraction	bounded side imbalance

OPEN PROVENANCE GATE

Exchange timestamps are not receive-time proof

The historical archives establish exchange event chronology, not when a live system received and accepted each event. A production study must retain receive time, source epoch, sequence, completeness decision, and acceptance-policy identity. This report does not claim that the archive alone proves live receive-time causality.

ENCODING

Exact-row representation and feature contract

Accepted bars keep every observed row; the 1,024-row ceiling is an explicit rejection bound.

IMPLEMENTED

Ragged storage without silent compaction

All row vectors are stored in one flat tensor with offsets. A window pads only to its largest observed bar and supplies a mask. Wide windows receive smaller batches under the row-token budget; no accepted row is ranked away, merged, or silently truncated.

IMPLEMENTED

Twenty-three bar features

The aggregate branch carries anchor-relative total and side POCs, centroids, low/high and value-area gaps, relative log volume, side and delta fractions, CVD change, stacked imbalance fractions, row count, feed quality, and cyclic minute/weekday coordinates.

Row group	Count	Meaning
Price geometry	1	POC-relative row offset
Side shares	4	buy, sell, total, delta shares
Log side ratios	2	buy and sell activity
Auction flags	4	bid/ask imbalance and zero-side flags

Bar group	Count	Meaning
Price / value area	10	POCs, centroids, range, VA
Volume / flow	7	volume, sides, delta, CVD, stacks
Width / quality	2	row count and feed quality
Calendar	4	minute and weekday sin/cos

BOUNDARY

Exact rows does not mean unlimited rows

A bar with more than 1,024 observed rows fails the current schema. The frozen maximum is reported separately; acceptance on other symbols or volatility regimes is not inferred.

CONTINUITY

Every context-plus-label interval must be regular

A candidate window is accepted only when all 64 context bars and all eight target bars are present at the configured one-minute cadence. The POC anchor and recent median log-volume reference are computed only from the context.

METHOD

Information contract and architecture

Every arrow is causal; forecasts are parallel marginal quantiles, not a joint candle path.



Causality

UTC one-minute bars are exact and contiguous. Context-only POC and volume references are frozen before the first target interval.

Exact-row policy

Accepted bars retain every observed row up to the 1,024-row reject bound. Batching reduces item count for wide candles; it never compacts an accepted row.

Output boundary

Low and high contain total and side POCs quantilewise. Marginals do not define a joint price path, product-distribution delta quantile, or order instruction.

MODEL DERIVATION

Architecture I - row pooling and fusion

Learned summaries operate on the observed row set; semantic query names remain hypotheses.

IMPLEMENTED

Per-row encoder

An identical two-layer MLP projects each 11-feature row independently to model width, then RMS-normalizes it. Rows do not mix until cross-attention, and padded positions are excluded by the row mask.

ROW EMBEDDING

e[t,j] = RMSNorm(W2 * GELU(W1 * x[t,j] + b1) + b2)

same weights for every observed price row

DESIGN HYPOTHESIS

Four auction queries

Four learned queries are intended to summarize total, buy-side, sell-side, and imbalance structure. Attention pictures cannot prove those roles. Query swaps, feature permutation, interventions, and seed stability are required interpretation tests.

MASKED QUERY POOLING

alpha[k,j] = softmax_j((Wq*q[k]) dot (Wk*e[j]) / sqrt(d_head) + mask[j])

a[k] = sum_j alpha[k,j] * Wv*e[j]

score-count term is O(K*n_rows), with K = 4

IMPLEMENTED

Gated row/aggregate fusion

The four query outputs are flattened into one row-set vector. A separate projection embeds the 23 deterministic aggregates. A sigmoid feature gate mixes them, while a residual projection prevents the representation from being only a convex mixture.

FUSION

g = sigmoid(Wg * concat(a_bar, b_bar) + bg)

u = g * a_bar + (1-g) * b_bar + Wr * concat(a_bar, b_bar)

Required ablations: aggregates-only, rows-only, concatenation, addition, mean/max pooling, one query versus four queries, and matched-capacity alternatives.

MODEL DERIVATION

Architecture II - causal history and direct horizons

Dense attention is proportionate at 64 history positions; no generated forecast is recycled.

IMPLEMENTED

Causal temporal stack

Three pre-RMSNorm blocks process 64 fused bar tokens at width 64. Each block uses masked multi-head attention, SwiGLU, residual connections, and LayerScale. The triangular mask preserves an online interpretation for every intermediate token.

TEMPORAL BLOCK

RMSNorm(x) = gain * x / sqrt(mean(x^2) + epsilon)
Z = H + gamma_A * MHA(RMSNorm(H), causal_mask)
H_next = Z + gamma_F * SwiGLU(RMSNorm(Z))
mask[i,j] = 0 when j <= i; -infinity otherwise

IMPLEMENTED

Parallel horizon decoder

Eight learned horizon queries first self-attend to one another, then cross-attend to the history memory. Horizon attention has no causal triangle because every query contains only learned identity - never a target realization or an earlier prediction.

DIRECT, NOT AUTOREGRESSIVE

z0[h] = learned_query[h] + learned_horizon_embedding[h]
(Y_hat[1], ..., Y_hat[8]) = F_theta(X[t-63:t])
NOT Y_hat[h] = f(X, Y_hat[1:h-1])

RATIONALE

Why exact dense attention

With L=64, the temporal sequence has only 4,096 position pairs per layer and head. The row-query term is linear in observed rows, and the decoder has only eight queries. Sparse attention, recurrent long-memory state, KV-cache compression, and expert routing solve different scale problems and add unjustified complexity here.

ATTENTION SCORE COMPLEXITY

O(L*K*n_bar) + O(N_temporal*L^2)
+ O(N_decoder*H*L) + O(N_decoder*H^2)
L=64, K=4, H=8

OUTPUT CONTRACT

Direct targets, quantiles, and physical constraints

Five price marginals share one cutoff anchor; volume and flow use separate monotone coordinates.

IMPLEMENTED

Context-only normalization

Let p_anchor be the final context POC, S_p=64 row ticks, S_v=4, and logV_ref the median log-volume over at most the final 16 context bars. No future observation enters these references.

SEVEN DIRECT TARGETS

price target k: y[k,h] = clip((p[k,t+h]-p_anchor)/S_p, -2, 2)

volume: y[V,h] = clip((log(1+V[t+h])-logV_ref)/S_v, -2, 2)

flow: y[Delta,h] = clip((V+ - V-)/(V+ + V-), -1, 1)

WHY DIRECT

No composition of uncertain price forecasts

Low, high, total POC, buy POC, and sell POC are all forecast directly from the same cutoff POC. The model does not predict a future POC and then add a separately uncertain gap to create another target.

IMPLEMENTED

Ordered q10 / q50 / q90

For each horizon and target, the head predicts a center plus positive softplus spreads. This prevents within-target quantile crossing before physical price constraints are applied.

NON-CROSSING PARAMETERIZATION

Q50 = center

Q10 = center - softplus(a) - 1e-4

Q90 = center + softplus(b) + 1e-4

QUANTILEWISE PRICE CONSTRAINTS

low = min(raw_low, total_POC)

high = max(raw_high, total_POC)

buy_POC = clamp(raw_buy_POC, low, high)

sell_POC = clamp(raw_sell_POC, low, high)

The operations preserve quantile order and guarantee low <= every POC <= high at each quantile. Delta-fraction quantiles pass through tanh. Structural order does not imply calibration, and constraint intervention rates must be measured.

PROBABILITY

Pinball loss and the marginal-versus-joint boundary

Proper marginal scores answer a narrower question than coherent candle-path probability.

IMPLEMENTED

Pinball loss

For residual e = y - q_hat at quantile tau, underprediction above the target quantile is penalized differently from overprediction. At q90, one unit of underprediction costs nine times one unit of overprediction.

CRITICAL LIMIT

Marginal quantiles are not a joint candle

FDT-N estimates seven conditional marginals. Even when each 80% interval is calibrated, their Cartesian product is not an 80% joint region. Quantilewise price ordering is a physical constraint, not a learned dependence model.

WEIGHTED MASKED OBJECTIVE

rho_tau(e) = max(tau*e, (tau-1)*e)

Loss = sum(mask * horizon_weight * target_weight * rho) / sum(weights)

horizon_weight[h] = 1/sqrt(h)

target weights = (1.0, .7, .7, .8, .8, .5, .7)

MONOTONE PHYSICAL INVERSES

price[k,h,tau] = p_anchor + S_p * row_size * Q[k,h,tau]

volume[h,tau] = exp(logV_ref + S_v*Q[V,h,tau]) - 1

EVALUATION RULE

Training weights cannot hide weak cells

The paper publishes unweighted q10/q50/q90 pinball, q50-forecast MAE and bias, coverage, and width for every horizon-target cell and every declared seed.

NOT IMPLEMENTED AS A QUANTILE

Physical delta requires joint information

q50(volume) times q50(delta fraction) is only a plug-in point diagnostic. Multiplying same-quantile endpoints does not create a calibrated delta interval. A direct delta head, joint model, or separately validated conservative map is required.

BEGINNER WORKED EXAMPLE

Worked footprint and forecast example

All numbers on this page are illustrative arithmetic, not model results.

Price	Buy	Sell	Total	Delta	ZeroBid	ZeroAsk
100.50	8	0	8	+8	1	0
100.25	32	6	38	+26	0	0
100.00	20	30	50	-10	0	0
99.75	5	22	27	-17	0	0
99.50	0	7	7	-7	0	1

DERIVATION

Bar summaries

V+ = 65, V- = 65, V = 130, and Delta = 0. Total POC is 100.00, buy POC is 100.25, and sell POC is 100.00. The volume-weighted total centroid is 100.025. Starting at POC volume 50, adding the 38-volume row above and 27-volume row below reaches 115, which exceeds the 70% value-area target of 91.

ROW ENCODING

The 100.25 row

With row size 0.25, the row sits one tick above POC. Its buy, sell, total, and delta shares are 32/130=0.2462, 6/130=0.0462, 38/130=0.2923, and 26/130=0.2000.

ILLUSTRATIVE FORECAST

Direct median decoding

From cutoff POC 100.00, offsets of POC +1, low -1, high +4, buy POC +1, and sell POC 0 decode to 100.25, 99.75, 101.00, 100.25, and 100.00. If the context volume reference is 100 and the relative log-volume median is 0.2, volume is 101*exp(0.2)-1 = 122.36.

PINBALL ARITHMETIC

One realized POC

For illustrative q10/q50/q90 POC offsets (-1, 1, 3) and realization 2 ticks, pinball terms are 0.3, 0.5, and 0.1, averaging 0.3 tick. A delta-fraction median of 0.10 times 122.36 gives plug-in delta 12.24; it is not a delta quantile.

EXPERIMENT DESIGN

Training, selection, and frozen protocol

All declared seeds share one budget; only validation loss selects the primary run.

Setting	Value	Evidence status
Optimizer	AdamW	implemented
Peak learning rate	3.00e-04	implemented
Weight decay	1.00e-03	implemented
Warm-up	8.0% of steps	implemented
Gradient clip	1.00	implemented
Epoch budget	30	same for every seed
Batch item cap	128	same for every seed
Patience	8	validation only
Row-token reference	64	exact rows retained

IMPLEMENTED

Purged chronological split

Windows remain chronological: 70% train, 15% validation, 15% test, with 72-bar purges around boundaries. Standardization and direct-ridge fitting use training windows only. Random window shuffling never defines a partition.

SELECTION RULE

No held-out test selection

The primary is seed 17, selected by minimum validation loss (0.112634); ties break by seed number. Every seed must share the same model, data, split, and non-seed training configuration. Every seed's held-out metrics remain visible regardless of rank.

REQUIRED FOR FINAL QUALIFICATION

Freeze before opening the outer test

Freeze sources, hashes, venue, instrument, row size, classification, value area, quality, gap handling, wall-clock boundaries, model, optimization, loss, calibration, and all baselines. Preserve at least three declared seeds. Open a later outer holdout once, then publish every target-horizon cell, missing-label count, clipping rate, and failure.

This report validates the declared multi-seed bundle and selected deployment artifact, but one 14-day corpus and one chronological split remain development evidence, not broad qualification.

COMPARATORS

Baselines and ablations

Implemented point comparators are reported honestly; richer distributional controls remain required.

Family	Output	Fairness condition	Status
Persistence	causal point	same test windows	implemented
Direct ridge	causal point	train-only fit	implemented
Rolling / seasonal	point	time-aware	required
Quantile linear	distribution	matched labels	required
Boosted trees	distribution	aggregate inputs	required
Aggregate MLP	distribution	matched budget	required
GRU / TCN	distribution	causal sequence	required
DeepLOB-like	distribution	structural control	required
Time-series transformer	distribution	causal adapter	required

REQUIRED / PROPOSED

Ablations that test the architecture claims

Aggregate-only versus full rows; row-only versus gated fusion; one/four queries versus mean/max pooling; feature-group removals; causal transformer versus TCN/GRU; learned positions versus relative age; parallel decoder versus independent heads; equal versus 1/sqrt(h) weights; lookbacks 16/32/64/128; and matched parameter-capacity controls.

BOUNDARY

Point baselines do not test calibration

Persistence and ridge compare model medians with point estimates. They cannot establish whether q10/q90 are sharp or calibrated. A claim about distribution quality needs task-matched quantile baselines and proper scores.

TRANSFER BOUNDARY

Prior negative evidence - CHMT-Q

A cautionary development study, not a matched FDT-N baseline and not reusable data.

OBSERVED PRIOR RESULT

What CHMT-Q actually studied

CHMT-Q used aggregate one-minute bars, half-second L1 tokens, fixed-depth L2 snapshots, and quality features on an OKX BTC-USDT period. It did not consume FDT-N's ragged footprint rows, used five horizons and five targets, and had a different coordinate, source contract, dates, and restricted dataset.

DECISION

Do not deploy; allocate no capital

CHMT-Q's preregistered uncertainty policy made zero trades. The transferable lesson is narrow: a modest proper-score improvement does not imply useful point accuracy, candle direction, trades, or economic value. Its scores must never be shown beside FDT-N as if both models saw the same inputs, labels, venue, or period.

worse H1-H3

POC Q50 MAE

tied H4-H5

52.67%

DIRECTION PROXY

MCC 0.0475

0

QUALIFIED TRADES

all POC intervals crossed 0

sealed

OUTER HOLDOUT

2026-06-25 to 06-30

The restricted CHMT-Q data and sealed outer holdout are neither opened nor copied for this study. The local CHMT-Q report is cited as prior negative evidence only.

METHODS

Evaluation definitions and dependence treatment

Proper scores, calibration, width, and point errors answer different questions.

PRIMARY DISTRIBUTIONAL METRICS

Every horizon-target cell

Report q10, q50, and q90 pinball; empirical quantile calibration; central-80% coverage; mean interval width; and, where implemented, interval score. Coverage without width can reward diffuse intervals, while width without coverage can reward unsafe narrowness.

CENTRAL INTERVAL SCORE (alpha = 0.20)

IS = (upper-lower)
+ (2/alpha)*(lower-y) when y < lower
+ (2/alpha)*(y-upper) when y > upper

SECONDARY METRICS

Point and domain diagnostics

Q50-forecast MAE and signed bias; total-POC direction as a secondary diagnostic; side-label availability; invariant violations; missing outputs; and conditional slices by horizon, volume, delta, volatility, session, quality, clipping, venue, and regime.

DEPENDENCE-AWARE DESCRIPTION

Paired four-hour block sensitivity intervals

Model-minus-baseline absolute errors stay paired within 13 contiguous 240-example blocks and are resampled 2,000 times. This preserves within-block structure better than IID resampling.

RIGOR LIMIT

Adjacent boundary windows still overlap

Forecast windows on either side of a four-hour block boundary share context and target bars. The blocks are therefore not independent. The displayed ranges are descriptive sensitivity intervals - not independent-block confidence intervals, causal effects, p-values, or formal generalization bounds.

LATER CONFIRMATORY STUDY

Predeclare the dependence unit

Choose the resampling or loss-differential unit before opening the outer test, justify it from sampling diagnostics, state any horizon correction and multiplicity policy, and publish effect sizes and intervals rather than a winner label alone.

DATA

Frozen corpus, split, and provenance

Manifest-identified, validated, and evaluated through a frozen, validation-only

20,160	357,676	20,089	605
ONE-MINUTE BARS	EXACT PRICE ROWS	VALID 64+8 WINDOWS	MAXIMUM ROWS/BAR
14 contiguous days	p50 13 | p99 94.41	before purged split	p95 47 | bound 1,024



72-bar purge on each boundary | no calibration partition

SOURCE	Binance Data Vision spot daily individual trades; live @trade granularity
RAW PROVENANCE	14 external archives; 374,843,570 original bytes; hashes in manifest
LOCAL RAW STATE	ZIP copies truncated after encoding; raw bytes recoverable, not locally retained
ROW/QUANTITY	2.5 USDT rows; SizeScale=1000; ties-to-even; zero scaled trades dropped
MANIFEST SHA-256	b88d89565febb97dbe56151fe447e3952fa844b8c433ecb86cb747e11e4ce412
DATASET SHA-256	d83f78894af08b65123d2bf9f4fbef8ca28e901a552d3af2c61f6e37513bf6cc

Input clipping and row-bound saturation

Raw bar coordinates are tested before clipping; the already encoded row coordinate has a narrower audit.

Input	Feature	Low	Low hit rate	High	High hit rate	N	Method
row	price offset from poc	-2	1.4222%	2	2.0840%	357,676	bound saturation
bar	poc offset from anchor	-2	4.0210%	2	3.8880%	1,285,696	raw pre-clip
bar	buy poc offset	-2	0.0199%	2	0.0647%	1,285,696	raw pre-clip
bar	sell poc offset	-2	0.0398%	2	0.0398%	1,285,696	raw pre-clip
bar	volume centroid offset	-2	0.0199%	2	0.0747%	1,285,696	raw pre-clip
bar	buy centroid offset	-2	0.0299%	2	0.0747%	1,285,696	raw pre-clip
bar	sell centroid offset	-2	0.0249%	2	0.0647%	1,285,696	raw pre-clip
bar	low gap	-	-	2	0.7268%	1,285,696	raw pre-clip
bar	high gap	-	-	2	0.6322%	1,285,696	raw pre-clip
bar	value area low gap	-	-	2	0.1294%	1,285,696	raw pre-clip
bar	value area high gap	-	-	2	0.2190%	1,285,696	raw pre-clip
bar	log volume relative	-2	0.0000%	2	0.0000%	1,285,696	raw pre-clip
bar	cumulative delta change	-2	0.0000%	2	0.0000%	1,285,696	raw pre-clip
bar	stacked buy fraction	-	-	1	0.0000%	1,285,696	raw pre-clip
bar	stacked sell fraction	-	-	1	0.0000%	1,285,696	raw pre-clip
bar	row count fraction	-	-	1	0.0000%	1,285,696	raw pre-clip

A raw pre-clip hit uses a strict inequality, so a value exactly equal to the bound is not counted as lost tail information. The frozen NPZ does not retain the row price before its initial clip; its row rate therefore counts encoded values at either boundary and may include exact-bound observations. Exact row pre-clip loss is explicitly unimplemented.

TAIL-LOSS AUDIT

Target clipping audit

Every valid horizon-target label is evaluated in its raw coordinate before the training clip.

Target	Low	Low clipped	High	High clipped	Valid labels
Total POC	-2	0.6104%	2	0.4860%	160,712
Low	-2	0.9607%	2	0.2047%	160,712
High	-2	0.2576%	2	0.7834%	160,712
Buy POC	-2	0.5949%	2	0.4922%	160,712
Sell POC	-2	0.6285%	2	0.4872%	160,712
Log volume	-2	0.0000%	2	0.0000%	160,712
Delta fraction	-1	0.0000%	1	0.0000%	160,712

MEASURED

Largest single target-bound fraction

Low reaches 0.9607% at its more active bound. The table gives every lower and upper fraction separately; no pooled average hides a target or direction.

DENOMINATOR

All continuous valid windows

Each price, volume, and delta target is counted over every valid horizon label. Buy-POC and sell-POC denominators exclude labels with no relevant side volume, matching the target mask.

INTERPRETATION

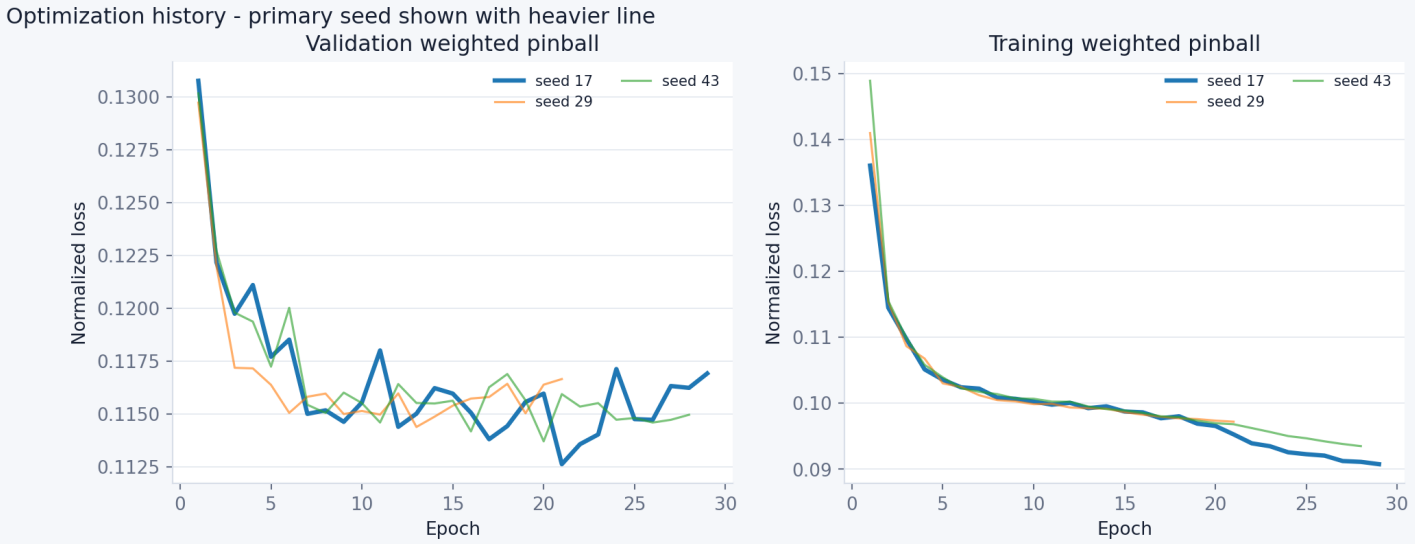
Descriptive corpus audit

Overlapping windows repeat bars, so these fractions describe the model's training/evaluation coordinate exposure. They are not independent statistical observations or a robustness proof.

TRAINING

Optimization history

Validation traces are selection evidence; held-out test metrics remain outside checkpoint choice.



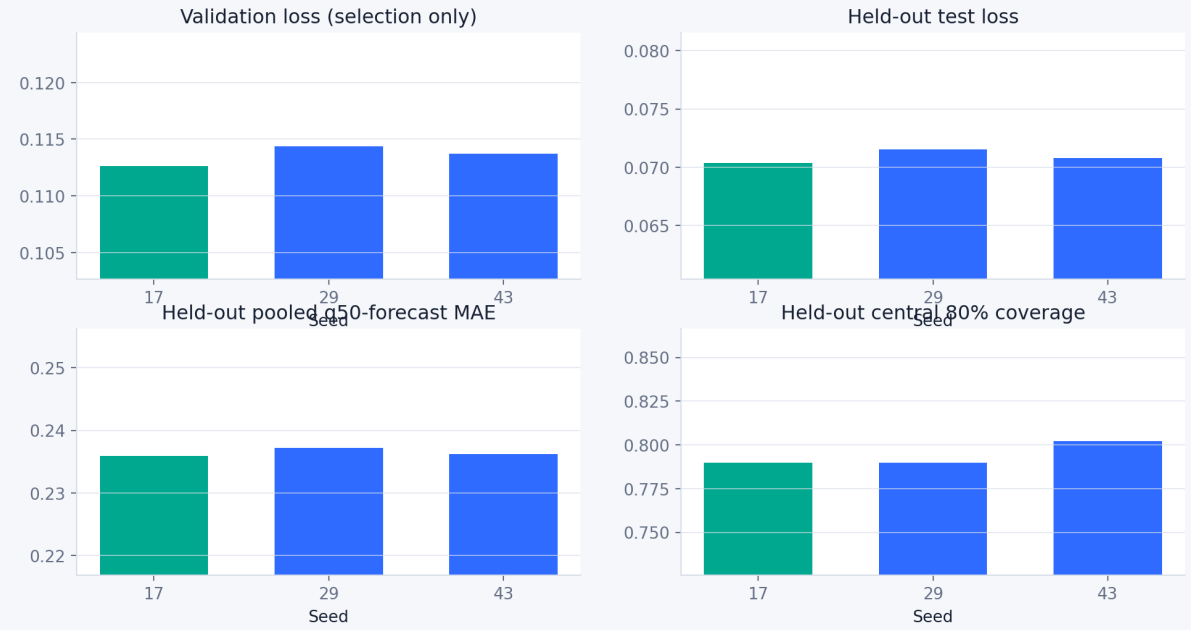
PRIMARY SEED	17
SELECTION	validation loss only
DEVICE CLASS	cuda
ELAPSED	1.783 hours
BEST EPOCH	21
EPOCHS COMPLETED	29
PEAK LEARNING RATE	3.00e-04
WEIGHT DECAY	1.00e-03
BATCH ITEM CAP	128
ROW-TOKEN REFERENCE	64
BOOTSTRAP RESAMPLES	2,000
D_MODEL / HEADS	64 / 4
PARAMETERS	371,861

MULTI-SEED

Seed stability

Primary selection uses validation loss only; all held-out summaries remain visible.

Every declared seed - teal marks validation-selected primary



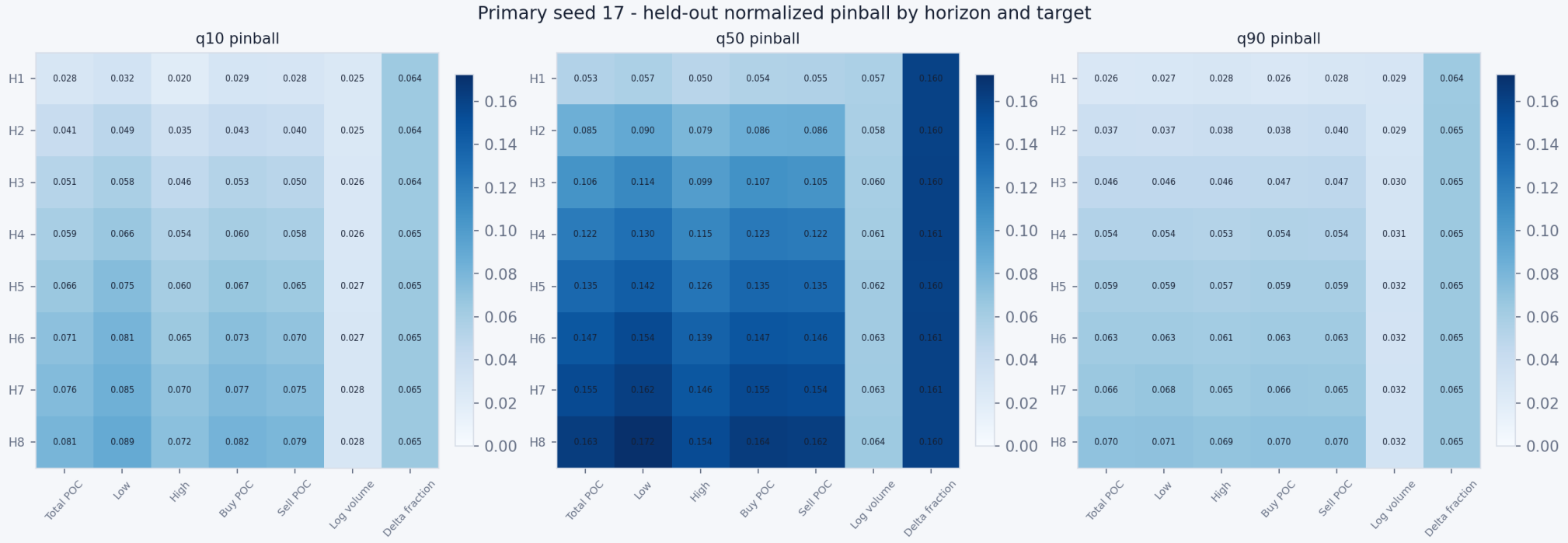
Seed	Status	Val loss	Test loss	Test MAE	80% cov
17	PRIMARY	0.11263	0.07039	0.2359	79.0%
29	reported	0.11439	0.07154	0.2372	79.0%
43	reported	0.11371	0.07079	0.2362	80.2%

Stability summary: test loss mean 0.0709, sd 0.0005 | pooled MAE mean 0.2364, sd 0.0006 | coverage mean 0.7939, sd 0.0059

HELD-OUT TEST

Primary probabilistic scores

Every horizon-target-quantile cell is reported in normalized target space; lower pinball is better.

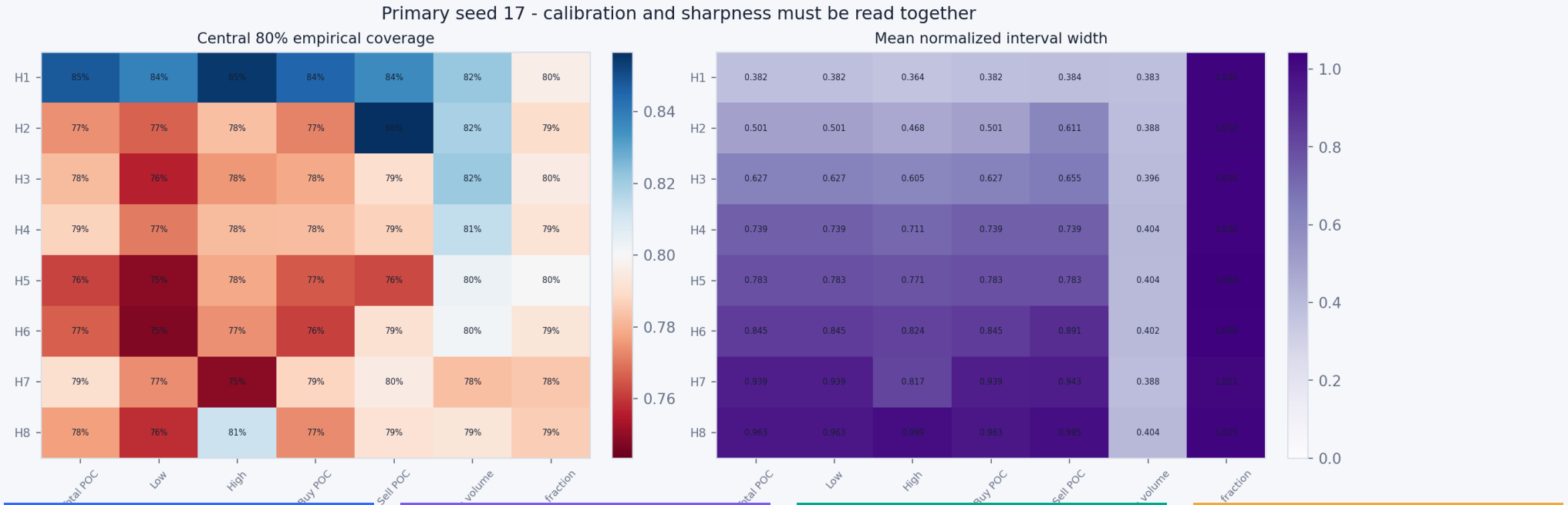


Primary seed 17 pooled pinball: q10 0.05549, q50 0.11793, q90 0.05139. These are marginal distribution scores, not a coherent joint path or economic result.

HELD-OUT TEST

Calibration and sharpness

Coverage without width can reward diffuse intervals; width without coverage can reward unsafe narrowness.



79.0%

POOLED COVERAGE

nominal 80%

0.7201

POOLED WIDTH

normalized

0.2359

POOLED Q50 MAE

normalized

54.9%

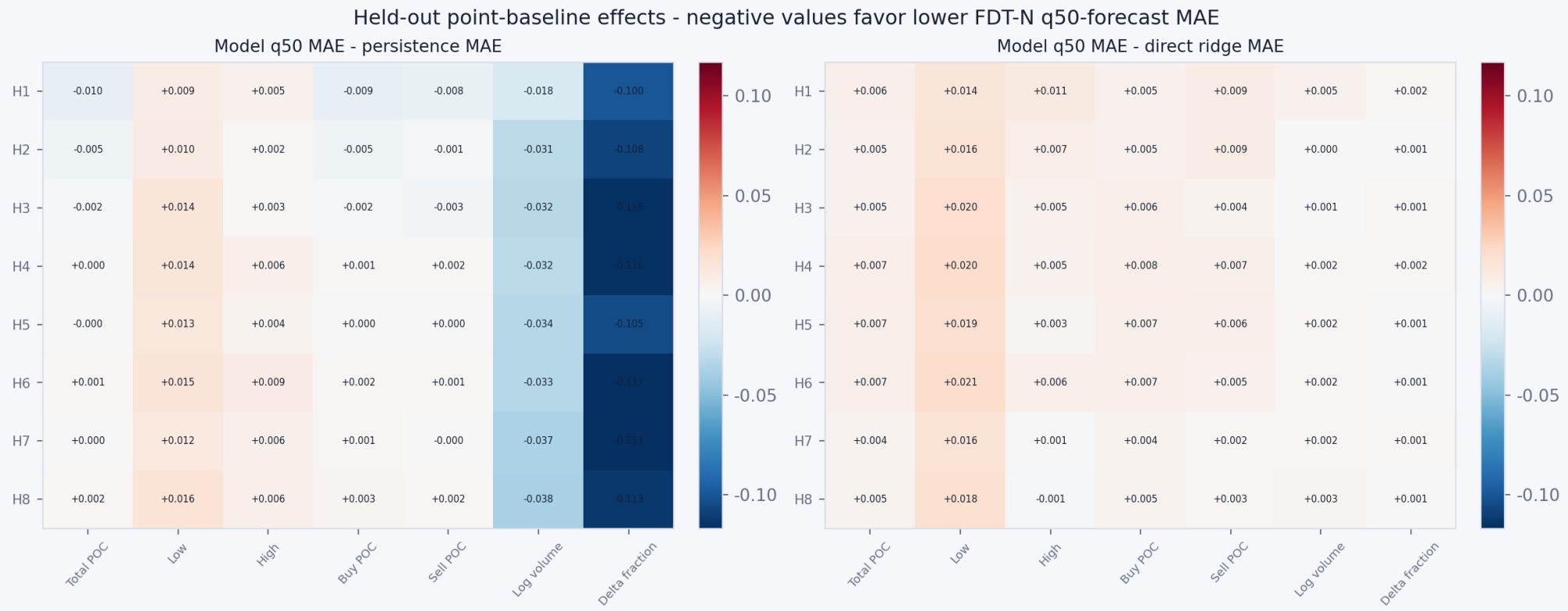
POC SIGN HIT

secondary metric

PAIRED EFFECTS

Point-baseline comparisons

Persistence and direct ridge are point baselines; they cannot establish quantile calibration.

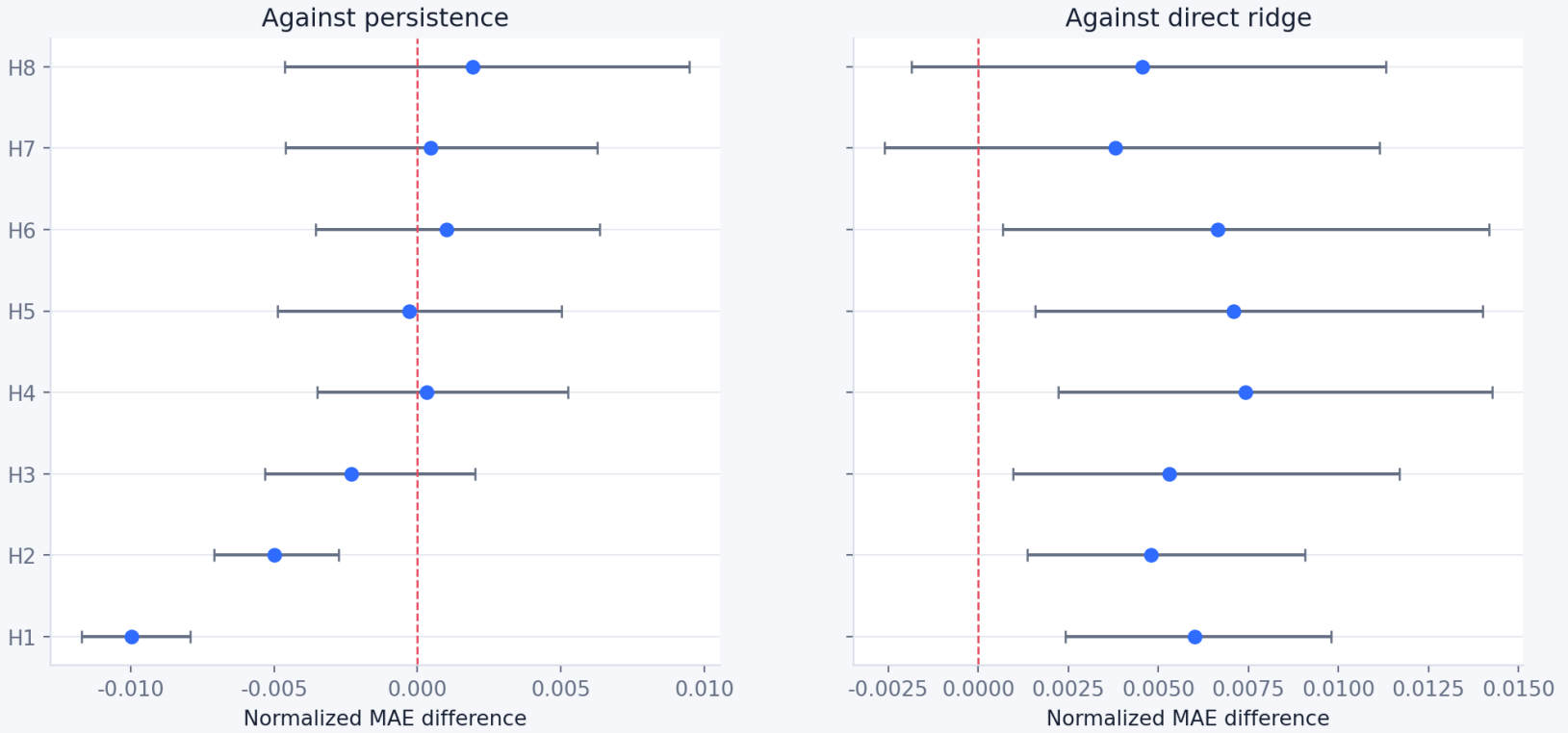


Effects are model q50-forecast MAE minus baseline MAE. Negative cells favor FDT-N. Intervals use 2,000 paired whole-block resamples over 13 contiguous four-hour blocks. Adjacent block-boundary windows still overlap, so these are descriptive sensitivity intervals. Cell multiplicity is descriptive.

Total-POC effect intervals

Paired four-hour sensitivity intervals preserve within-block dependence; adjacent boundary windows overlap.

Primary seed 17 total-POC paired four-hour sensitivity intervals



The dashed zero line marks equal q50-forecast MAE. An interval crossing zero is compatible with either ordering under this descriptive resampling design. The intervals are not independent-block or formal generalization bounds. No p-value or trading inference is made.

PRIMARY RUN

Held-out scorecard - seed 17

0.11263	0.07039	54.9%	PRIMARY
VALIDATION LOSS	HELD-OUT TEST LOSS	POC SIGN HIT	RUN STATUS
selection input	post-selection	secondary	validation-only rule

Target	q10 PB	q50 PB	q90 PB	Q50 MAE	Bias	80% cov	Width
Total POC	0.0592	0.1207	0.0526	0.2414	+0.0268	78.6%	0.7224
Low	0.0669	0.1276	0.0530	0.2552	+0.1183	77.0%	0.7224
High	0.0527	0.1136	0.0521	0.2272	+0.0290	78.9%	0.6948
Buy POC	0.0604	0.1216	0.0529	0.2431	+0.0418	78.3%	0.7224
Sell POC	0.0582	0.1207	0.0533	0.2415	+0.0129	80.1%	0.7502
Log volume	0.0264	0.0609	0.0309	0.1219	-0.0091	80.7%	0.3961
Delta fraction	0.0646	0.1604	0.0648	0.3207	+0.0239	79.2%	1.0324

PB = normalized pinball. Target summaries weight horizon cells by valid example count. The seven marginals are not a joint candle distribution.

PRIMARY RUN DETAIL

Seed 17 held-out cells (1/3)

Every horizon, target, and reported quantile score; no cell is hidden by pooling.

H	Target	N	q10 PB	q50 PB	q90 PB	Q50 MAE	Bias	80% cov	Width
1	Total POC	2,942	0.02820	0.05326	0.02595	0.10652	-0.01427	84.74%	0.38201
1	Low	2,942	0.03209	0.05682	0.02657	0.11365	+0.07805	83.82%	0.38201
1	High	2,942	0.01983	0.04999	0.02808	0.09998	+0.00496	85.45%	0.36447
1	Buy POC	2,942	0.02906	0.05423	0.02578	0.10847	+0.00078	84.47%	0.38201
1	Sell POC	2,942	0.02839	0.05504	0.02780	0.11009	-0.03094	83.62%	0.38363
1	Log volume	2,942	0.02458	0.05672	0.02881	0.11344	-0.00140	82.15%	0.38277
1	Delta fraction	2,942	0.06435	0.15979	0.06447	0.31958	+0.02126	79.61%	1.03181
2	Total POC	2,942	0.04133	0.08508	0.03745	0.17016	+0.00635	77.43%	0.50121
2	Low	2,942	0.04881	0.09009	0.03692	0.18019	+0.09820	76.65%	0.50122
2	High	2,942	0.03525	0.07921	0.03790	0.15843	+0.01562	78.31%	0.46798
2	Buy POC	2,942	0.04271	0.08568	0.03772	0.17136	+0.02134	77.19%	0.50122
2	Sell POC	2,942	0.04033	0.08628	0.03986	0.17257	+0.01317	85.66%	0.61107
2	Log volume	2,942	0.02542	0.05841	0.02945	0.11681	-0.00852	81.85%	0.38771
2	Delta fraction	2,942	0.06424	0.16021	0.06473	0.32042	+0.02137	79.03%	1.03030
3	Total POC	2,942	0.05124	0.10585	0.04642	0.21171	+0.02610	78.21%	0.62652
3	Low	2,942	0.05849	0.11374	0.04629	0.22748	+0.11766	75.59%	0.62653
3	High	2,942	0.04559	0.09886	0.04602	0.19772	+0.02626	77.53%	0.60462
3	Buy POC	2,942	0.05263	0.10730	0.04692	0.21460	+0.04097	77.84%	0.62653
3	Sell POC	2,942	0.05038	0.10518	0.04722	0.21035	+0.00910	79.06%	0.65524

PB = normalized pinball. Coverage is empirical q10-to-q90 inclusion; width is in the encoded target coordinate. These are marginal forecast cells, not joint paths.

PRIMARY RUN DETAIL

Seed 17 held-out cells (2/3)

Every horizon, target, and reported quantile score; no cell is hidden by pooling.

H	Target	N	q10 PB	q50 PB	q90 PB	Q50 MAE	Bias	80% cov	Width
3	Log volume	2,942	0.02571	0.05970	0.03023	0.11940	-0.00157	82.09%	0.39597
3	Delta fraction	2,942	0.06444	0.16040	0.06481	0.32080	+0.02960	79.54%	1.03534
4	Total POC	2,942	0.05912	0.12220	0.05376	0.24441	+0.02262	78.76%	0.73924
4	Low	2,942	0.06606	0.12959	0.05365	0.25917	+0.11421	77.06%	0.73922
4	High	2,942	0.05403	0.11487	0.05282	0.22973	+0.02883	78.21%	0.71063
4	Buy POC	2,942	0.06025	0.12347	0.05417	0.24693	+0.03783	78.25%	0.73922
4	Sell POC	2,942	0.05836	0.12237	0.05364	0.24474	+0.00591	78.69%	0.73921
4	Log volume	2,942	0.02622	0.06083	0.03089	0.12167	-0.00806	81.41%	0.40430
4	Delta fraction	2,942	0.06459	0.16069	0.06474	0.32138	+0.02856	79.23%	1.03220
5	Total POC	2,942	0.06576	0.13482	0.05885	0.26963	+0.02640	76.17%	0.78321
5	Low	2,942	0.07491	0.14200	0.05861	0.28400	+0.11789	74.92%	0.78314
5	High	2,942	0.06008	0.12625	0.05746	0.25249	+0.01373	77.87%	0.77098
5	Buy POC	2,942	0.06695	0.13548	0.05907	0.27096	+0.04153	76.51%	0.78314
5	Sell POC	2,942	0.06453	0.13478	0.05898	0.26956	+0.00968	76.27%	0.78314
5	Log volume	2,942	0.02678	0.06200	0.03173	0.12400	-0.01003	80.35%	0.40365
5	Delta fraction	2,942	0.06470	0.16022	0.06494	0.32045	+0.01487	80.01%	1.04407
6	Total POC	2,942	0.07148	0.14664	0.06270	0.29327	+0.04022	76.61%	0.84536
6	Low	2,942	0.08120	0.15419	0.06328	0.30838	+0.13139	74.75%	0.84544
6	High	2,942	0.06484	0.13923	0.06137	0.27846	+0.06016	77.43%	0.82394

PB = normalized pinball. Coverage is empirical q10-to-q90 inclusion; width is in the encoded target coordinate. These are marginal forecast cells, not joint paths.

PRIMARY RUN DETAIL

Seed 17 held-out cells (3/3)

Every horizon, target, and reported quantile score; no cell is hidden by pooling.

H	Target	N	q10 PB	q50 PB	q90 PB	Q50 MAE	Bias	80% cov	Width
6	Buy POC	2,942	0.07280	0.14725	0.06304	0.29450	+0.05541	76.14%	0.84544
6	Sell POC	2,942	0.06999	0.14595	0.06300	0.29190	+0.02344	79.13%	0.89107
6	Log volume	2,942	0.02703	0.06258	0.03160	0.12516	-0.01202	80.15%	0.40177
6	Delta fraction	2,942	0.06493	0.16050	0.06476	0.32101	+0.02597	79.30%	1.03975
7	Total POC	2,942	0.07594	0.15471	0.06582	0.30942	+0.04687	79.10%	0.93864
7	Low	2,942	0.08454	0.16210	0.06772	0.32420	+0.13792	77.36%	0.93860
7	High	2,942	0.06975	0.14646	0.06452	0.29291	+0.04271	74.92%	0.81729
7	Buy POC	2,942	0.07705	0.15505	0.06627	0.31010	+0.06202	78.79%	0.93860
7	Sell POC	2,942	0.07464	0.15401	0.06550	0.30802	+0.03007	79.54%	0.94350
7	Log volume	2,942	0.02751	0.06324	0.03227	0.12649	-0.01451	78.31%	0.38821
7	Delta fraction	2,942	0.06482	0.16062	0.06495	0.32125	+0.02432	78.48%	1.02101
8	Total POC	2,942	0.08051	0.16320	0.07014	0.32640	+0.05988	77.70%	0.96324
8	Low	2,942	0.08937	0.17235	0.07133	0.34471	+0.15077	75.83%	0.96302
8	High	2,942	0.07246	0.15402	0.06902	0.30804	+0.04008	81.20%	0.99888
8	Buy POC	2,942	0.08158	0.16414	0.07033	0.32828	+0.07492	77.33%	0.96302
8	Sell POC	2,942	0.07924	0.16230	0.07031	0.32460	+0.04309	79.20%	0.99488
8	Log volume	2,942	0.02767	0.06397	0.03208	0.12794	-0.01645	79.30%	0.40430
8	Delta fraction	2,942	0.06484	0.16041	0.06492	0.32081	+0.02487	78.62%	1.02487

PB = normalized pinball. Coverage is empirical q10-to-q90 inclusion; width is in the encoded target coordinate. These are marginal forecast cells, not joint paths.

DECLARED SEED

Held-out scorecard - seed 29

0.11439	0.07154	49.4%	REPORTED
VALIDATION LOSS	HELD-OUT TEST LOSS	POC SIGN HIT	RUN STATUS
selection input	post-selection	secondary	validation-only rule

Target	q10 PB	q50 PB	q90 PB	Q50 MAE	Bias	80% cov	Width
Total POC	0.0605	0.1222	0.0530	0.2444	+0.0502	79.2%	0.7476
Low	0.0685	0.1273	0.0531	0.2547	+0.1045	75.6%	0.7152
High	0.0533	0.1141	0.0520	0.2283	-0.0130	78.9%	0.6975
Buy POC	0.0615	0.1234	0.0538	0.2467	+0.0561	79.5%	0.7631
Sell POC	0.0595	0.1219	0.0529	0.2437	+0.0416	80.8%	0.7706
Log volume	0.0265	0.0611	0.0307	0.1222	-0.0092	79.9%	0.3893
Delta fraction	0.0646	0.1604	0.0647	0.3207	+0.0160	78.8%	1.0238

PB = normalized pinball. Target summaries weight horizon cells by valid example count. The seven marginals are not a joint candle distribution.

DECLARED SEED DETAIL

Seed 29 held-out cells (1/3)

Every horizon, target, and reported quantile score; no cell is hidden by pooling.

H	Target	N	q10 PB	q50 PB	q90 PB	Q50 MAE	Bias	80% cov	Width
1	Total POC	2,942	0.03091	0.06000	0.02961	0.12000	+0.02644	82.90%	0.41024
1	Low	2,942	0.03849	0.06497	0.02800	0.12993	+0.07856	76.41%	0.36384
1	High	2,942	0.02368	0.05366	0.02856	0.10731	-0.02795	83.65%	0.36974
1	Buy POC	2,942	0.03233	0.06527	0.03272	0.13054	+0.05494	84.84%	0.47496
1	Sell POC	2,942	0.03072	0.06297	0.03168	0.12594	+0.03972	86.95%	0.48683
1	Log volume	2,942	0.02512	0.05757	0.02883	0.11514	-0.01701	83.00%	0.38942
1	Delta fraction	2,942	0.06430	0.16034	0.06448	0.32067	+0.02621	78.82%	1.01897
2	Total POC	2,942	0.04328	0.08795	0.03967	0.17589	+0.01526	78.76%	0.54751
2	Low	2,942	0.04973	0.09462	0.03950	0.18924	+0.10727	77.60%	0.54751
2	High	2,942	0.03668	0.08079	0.03930	0.16159	-0.00683	78.08%	0.48526
2	Buy POC	2,942	0.04420	0.08963	0.04039	0.17926	+0.03650	81.51%	0.59057
2	Sell POC	2,942	0.04261	0.08835	0.03971	0.17669	+0.03125	83.14%	0.60150
2	Log volume	2,942	0.02552	0.05896	0.02959	0.11792	-0.00529	80.83%	0.37976
2	Delta fraction	2,942	0.06444	0.16022	0.06464	0.32043	+0.01613	78.79%	1.01445
3	Total POC	2,942	0.05380	0.10959	0.04779	0.21917	+0.05631	78.38%	0.65201
3	Low	2,942	0.06140	0.11280	0.04721	0.22560	+0.08864	73.32%	0.61532
3	High	2,942	0.04608	0.09978	0.04701	0.19956	-0.01105	78.18%	0.59989
3	Buy POC	2,942	0.05431	0.11076	0.04869	0.22151	+0.06330	78.65%	0.67192
3	Sell POC	2,942	0.05190	0.10869	0.04709	0.21737	+0.05484	80.46%	0.67789

PB = normalized pinball. Coverage is empirical q10-to-q90 inclusion; width is in the encoded target coordinate. These are marginal forecast cells, not joint paths.

DECLARED SEED DETAIL

Seed 29 held-out cells (2/3)

Every horizon, target, and reported quantile score; no cell is hidden by pooling.

H	Target	N	q10 PB	q50 PB	q90 PB	Q50 MAE	Bias	80% cov	Width
3	Log volume	2,942	0.02609	0.06008	0.03009	0.12016	-0.01164	80.66%	0.38913
3	Delta fraction	2,942	0.06442	0.16021	0.06468	0.32042	+0.01117	79.40%	1.03218
4	Total POC	2,942	0.06062	0.12344	0.05324	0.24688	+0.05896	79.16%	0.75098
4	Low	2,942	0.06844	0.12722	0.05242	0.25444	+0.09037	73.59%	0.67957
4	High	2,942	0.05375	0.11456	0.05255	0.22911	-0.01520	77.97%	0.67786
4	Buy POC	2,942	0.06138	0.12328	0.05362	0.24656	+0.04945	78.35%	0.75217
4	Sell POC	2,942	0.05965	0.12182	0.05263	0.24364	+0.02628	79.88%	0.75655
4	Log volume	2,942	0.02648	0.06103	0.03075	0.12207	-0.00609	79.91%	0.38954
4	Delta fraction	2,942	0.06455	0.16014	0.06466	0.32028	+0.00694	79.03%	1.02441
5	Total POC	2,942	0.06699	0.13570	0.05783	0.27139	+0.06936	78.11%	0.81888
5	Low	2,942	0.07649	0.13983	0.05839	0.27966	+0.09627	74.10%	0.76617
5	High	2,942	0.05935	0.12660	0.05672	0.25319	-0.01045	78.28%	0.76645
5	Buy POC	2,942	0.06841	0.13593	0.05846	0.27187	+0.06437	77.53%	0.81962
5	Sell POC	2,942	0.06561	0.13454	0.05736	0.26908	+0.05647	78.69%	0.82288
5	Log volume	2,942	0.02684	0.06173	0.03097	0.12347	-0.00860	79.74%	0.39327
5	Delta fraction	2,942	0.06462	0.16025	0.06471	0.32050	+0.01261	78.08%	1.01437
6	Total POC	2,942	0.07196	0.14486	0.06200	0.28972	+0.05899	78.62%	0.87796
6	Low	2,942	0.08132	0.15326	0.06386	0.30652	+0.15020	76.78%	0.87764
6	High	2,942	0.06373	0.13826	0.06077	0.27652	-0.03165	80.15%	0.87728

PB = normalized pinball. Coverage is empirical q10-to-q90 inclusion; width is in the encoded target coordinate. These are marginal forecast cells, not joint paths.

DECLARED SEED DETAIL

Seed 29 held-out cells (3/3)

Every horizon, target, and reported quantile score; no cell is hidden by pooling.

H	Target	N	q10 PB	q50 PB	q90 PB	Q50 MAE	Bias	80% cov	Width
6	Buy POC	2,942	0.07318	0.14618	0.06233	0.29236	+0.07422	78.42%	0.87755
6	Sell POC	2,942	0.07064	0.14424	0.06138	0.28848	+0.04225	78.62%	0.87761
6	Log volume	2,942	0.02711	0.06247	0.03156	0.12494	-0.00699	79.81%	0.40110
6	Delta fraction	2,942	0.06471	0.16050	0.06464	0.32099	+0.01634	79.23%	1.03130
7	Total POC	2,942	0.07673	0.15451	0.06530	0.30902	+0.07120	78.93%	0.94066
7	Low	2,942	0.08483	0.15713	0.06580	0.31426	+0.08891	75.56%	0.88892
7	High	2,942	0.06942	0.14559	0.06398	0.29117	+0.00134	77.74%	0.88266
7	Buy POC	2,942	0.07802	0.15360	0.06562	0.30719	+0.04323	78.72%	0.93594
7	Sell POC	2,942	0.07564	0.15350	0.06491	0.30701	+0.04057	79.06%	0.93734
7	Log volume	2,942	0.02748	0.06324	0.03200	0.12649	-0.00758	77.36%	0.38207
7	Delta fraction	2,942	0.06473	0.16054	0.06477	0.32108	+0.02195	78.55%	1.02131
8	Total POC	2,942	0.07993	0.16141	0.06873	0.32282	+0.04535	78.76%	0.98297
8	Low	2,942	0.08756	0.16881	0.06965	0.33762	+0.13580	77.06%	0.98297
8	High	2,942	0.07333	0.15384	0.06708	0.30769	-0.00186	77.16%	0.92105
8	Buy POC	2,942	0.08042	0.16230	0.06894	0.32459	+0.06306	78.35%	0.98186
8	Sell POC	2,942	0.07914	0.16069	0.06849	0.32139	+0.04119	79.47%	1.00415
8	Log volume	2,942	0.02770	0.06385	0.03202	0.12771	-0.01070	78.11%	0.39030
8	Delta fraction	2,942	0.06505	0.16063	0.06499	0.32125	+0.01626	78.79%	1.03331

PB = normalized pinball. Coverage is empirical q10-to-q90 inclusion; width is in the encoded target coordinate. These are marginal forecast cells, not joint paths.

DECLARED SEED

Held-out scorecard - seed 43

0.11371	0.07079	50.7%	REPORTED
VALIDATION LOSS	HELD-OUT TEST LOSS	POC SIGN HIT	RUN STATUS
selection input	post-selection	secondary	validation-only rule

Target	q10 PB	q50 PB	q90 PB	Q50 MAE	Bias	80% cov	Width
Total POC	0.0598	0.1213	0.0520	0.2426	+0.0101	78.6%	0.7511
Low	0.0649	0.1257	0.0519	0.2515	+0.0979	78.6%	0.7594
High	0.0534	0.1141	0.0508	0.2282	-0.0131	81.8%	0.7510
Buy POC	0.0604	0.1225	0.0523	0.2451	+0.0339	79.6%	0.7670
Sell POC	0.0593	0.1220	0.0526	0.2441	+0.0373	81.5%	0.7940
Log volume	0.0264	0.0611	0.0309	0.1221	-0.0317	81.6%	0.4015
Delta fraction	0.0642	0.1598	0.0649	0.3197	+0.0157	79.9%	1.0382

PB = normalized pinball. Target summaries weight horizon cells by valid example count. The seven marginals are not a joint candle distribution.

DECLARED SEED DETAIL

Seed 43 held-out cells (1/3)

Every horizon, target, and reported quantile score; no cell is hidden by pooling.

H	Target	N	q10 PB	q50 PB	q90 PB	Q50 MAE	Bias	80% cov	Width
1	Total POC	2,942	0.03073	0.05956	0.02864	0.11912	-0.01632	78.76%	0.37634
1	Low	2,942	0.03505	0.06108	0.02679	0.12215	+0.06388	81.24%	0.39629
1	High	2,942	0.02415	0.05325	0.02794	0.10649	-0.01561	86.10%	0.39401
1	Buy POC	2,942	0.03146	0.06412	0.02895	0.12825	+0.05997	84.60%	0.43078
1	Sell POC	2,942	0.02946	0.06232	0.03090	0.12464	+0.03568	84.13%	0.43495
1	Log volume	2,942	0.02534	0.05715	0.02877	0.11429	-0.02841	83.68%	0.39816
1	Delta fraction	2,942	0.06383	0.15961	0.06460	0.31922	+0.00881	79.84%	1.03289
2	Total POC	2,942	0.04342	0.08882	0.03914	0.17765	-0.00540	77.53%	0.53676
2	Low	2,942	0.04906	0.09196	0.03804	0.18392	+0.08661	77.23%	0.53753
2	High	2,942	0.03674	0.08060	0.03799	0.16120	-0.00845	78.86%	0.50203
2	Buy POC	2,942	0.04474	0.08942	0.03911	0.17885	+0.00975	77.12%	0.53752
2	Sell POC	2,942	0.04284	0.08917	0.03976	0.17834	+0.04244	80.59%	0.57172
2	Log volume	2,942	0.02580	0.05878	0.02981	0.11756	-0.03245	81.82%	0.39132
2	Delta fraction	2,942	0.06396	0.15990	0.06487	0.31980	+0.02149	80.42%	1.04008
3	Total POC	2,942	0.05277	0.10749	0.04667	0.21498	+0.00990	78.11%	0.65620
3	Low	2,942	0.05763	0.11300	0.04589	0.22599	+0.09886	77.57%	0.65867
3	High	2,942	0.04638	0.09943	0.04563	0.19886	-0.01208	81.48%	0.65113
3	Buy POC	2,942	0.05329	0.10892	0.04685	0.21784	+0.02217	77.87%	0.65867
3	Sell POC	2,942	0.05234	0.10803	0.04701	0.21605	+0.03035	83.21%	0.74015

PB = normalized pinball. Coverage is empirical q10-to-q90 inclusion; width is in the encoded target coordinate. These are marginal forecast cells, not joint paths.

DECLARED SEED DETAIL

Seed 43 held-out cells (2/3)

Every horizon, target, and reported quantile score; no cell is hidden by pooling.

H	Target	N	q10 PB	q50 PB	q90 PB	Q50 MAE	Bias	80% cov	Width
3	Log volume	2,942	0.02590	0.05991	0.03012	0.11982	-0.03117	82.29%	0.39810
3	Delta fraction	2,942	0.06401	0.15981	0.06490	0.31962	+0.01530	79.74%	1.02897
4	Total POC	2,942	0.06003	0.12189	0.05230	0.24378	-0.00029	76.85%	0.73218
4	Low	2,942	0.06543	0.12633	0.05171	0.25266	+0.08933	76.89%	0.73322
4	High	2,942	0.05358	0.11389	0.05041	0.22778	+0.00313	80.35%	0.73375
4	Buy POC	2,942	0.06074	0.12290	0.05249	0.24580	+0.01295	76.72%	0.73322
4	Sell POC	2,942	0.05967	0.12307	0.05245	0.24613	+0.04125	82.49%	0.82096
4	Log volume	2,942	0.02621	0.06059	0.03091	0.12118	-0.02828	80.73%	0.39223
4	Delta fraction	2,942	0.06409	0.15979	0.06496	0.31957	+0.01202	80.18%	1.04405
5	Total POC	2,942	0.06529	0.13379	0.05647	0.26757	+0.01306	78.93%	0.82918
5	Low	2,942	0.07045	0.13887	0.05647	0.27775	+0.10093	78.25%	0.82949
5	High	2,942	0.05927	0.12677	0.05487	0.25354	-0.02062	80.52%	0.81042
5	Buy POC	2,942	0.06593	0.13482	0.05745	0.26964	+0.04489	80.86%	0.87049
5	Sell POC	2,942	0.06524	0.13457	0.05691	0.26915	+0.02632	80.59%	0.86993
5	Log volume	2,942	0.02671	0.06164	0.03121	0.12329	-0.02854	81.88%	0.40785
5	Delta fraction	2,942	0.06412	0.15973	0.06499	0.31946	+0.01187	80.29%	1.04627
6	Total POC	2,942	0.07083	0.14446	0.06033	0.28892	+0.02268	76.92%	0.85422
6	Low	2,942	0.07664	0.14936	0.06084	0.29872	+0.10941	78.52%	0.89489
6	High	2,942	0.06436	0.13712	0.05925	0.27424	-0.00853	84.60%	0.97039

PB = normalized pinball. Coverage is empirical q10-to-q90 inclusion; width is in the encoded target coordinate. These are marginal forecast cells, not joint paths.

DECLARED SEED DETAIL

Seed 43 held-out cells (3/3)

Every horizon, target, and reported quantile score; no cell is hidden by pooling.

H	Target	N	q10 PB	q50 PB	q90 PB	Q50 MAE	Bias	80% cov	Width
6	Buy POC	2,942	0.07154	0.14519	0.06051	0.29037	+0.03417	77.43%	0.86400
6	Sell POC	2,942	0.06985	0.14539	0.06106	0.29078	+0.05155	80.90%	0.92498
6	Log volume	2,942	0.02686	0.06270	0.03184	0.12540	-0.03303	80.69%	0.40504
6	Delta fraction	2,942	0.06430	0.15989	0.06475	0.31979	+0.01050	79.64%	1.03198
7	Total POC	2,942	0.07560	0.15314	0.06438	0.30627	+0.03735	80.56%	0.98878
7	Low	2,942	0.08064	0.15913	0.06605	0.31826	+0.12347	79.98%	0.99007
7	High	2,942	0.06960	0.14836	0.06363	0.29672	-0.05292	78.89%	0.89927
7	Buy POC	2,942	0.07607	0.15366	0.06476	0.30732	+0.04757	80.42%	0.99007
7	Sell POC	2,942	0.07564	0.15262	0.06434	0.30524	+0.02071	79.40%	0.96574
7	Log volume	2,942	0.02698	0.06370	0.03216	0.12741	-0.03645	80.12%	0.40199
7	Delta fraction	2,942	0.06461	0.16016	0.06492	0.32031	+0.02718	80.15%	1.04515
8	Total POC	2,942	0.07935	0.16112	0.06822	0.32224	+0.01964	81.07%	1.03491
8	Low	2,942	0.08418	0.16620	0.06959	0.33240	+0.11073	79.47%	1.03491
8	High	2,942	0.07305	0.15341	0.06639	0.30681	+0.01036	83.58%	1.04721
8	Buy POC	2,942	0.07957	0.16121	0.06867	0.32242	+0.03948	81.37%	1.05115
8	Sell POC	2,942	0.07958	0.16105	0.06827	0.32209	+0.04974	80.52%	1.02355
8	Log volume	2,942	0.02754	0.06405	0.03224	0.12811	-0.03566	81.65%	0.41758
8	Delta fraction	2,942	0.06456	0.15991	0.06489	0.31981	+0.01847	79.30%	1.03587

PB = normalized pinball. Coverage is empirical q10-to-q90 inclusion; width is in the encoded target coordinate. These are marginal forecast cells, not joint paths.

DEPLOYMENT ARTIFACT

Selected model artifact and runtime evidence

Metadata lineage binds the validation-selected results, checkpoint, ONNX graph, parity, and coordinate.

17	21	2.38e-07	14.641 ms
SELECTED SEED	CHECKPOINT EPOCH	PYTORCH VS ORT MAX ERROR	CPU LATENCY P95
validation-only	selected validation epoch	tolerance 1.0e-04	50 timed runs

RUNTIME STATUS	shadow-only-research
CREATES ORDER SIGNALS	false
COORDINATE	Binance / BTCUSDT
INTERVAL / ROW SIZE	60 seconds / 2.5 USDT
HISTORY / HORIZON	64 / 8 bars
REPRESENTATIVE WIDTH	187 rows (p95 187.00)
LATENCY P50 / P95 / P99	11.410 / 14.641 / 17.240 ms
ONNX RUNTIME PROVIDER	CPUExecutionProvider
PYTHON / PYTORCH	3.12.6 / 2.5.1+cu121
ONNX / ORT	1.22.0 / 1.28.0

CHECKPOINT SHA-256	9d2cf525971f81fc93ca9ed97103cffb29203d7fd59edea5bd92d0e14...
ONNX SHA-256	adce8cd39312bf3ad1389a23ed5ca24d6f47a7f58dbb26d76ae7d54a...
PRIMARY RESULTS SHA-256	746403a05790753943b8c22f615f097c8dc397e413be49f35ea553b8...
METADATA SHA-256	9f998ac6b56b008ac02aaa0ed5c4f08447b1735f33027294773d9e1a...

The runtime coordinate uses the product broker identity 'Binance'. Training provenance is separately frozen as Binance Data Vision spot daily individual trades. Passed fixture parity and representative CPU timing qualify this exact exported artifact for shadow-only research; they do not establish live end-to-end latency, drift resistance, or trading safety.

PRODUCT SAFETY

Runtime contract, fail-closed behavior, and governance

Forecast display must never become an order path; native footprint rendering survives abstention.

IMPLEMENTED SHARED SEAM

Explicit coordinate and status

Requests bind instrument, source, interval, row size, cutoff, and target interval. Results distinguish Available, Unavailable, InsufficientHistory, and Failed. Available values carry provider/model/version identity and ordered q10/q50/q90 for all seven target families.

PASSED ARTIFACT GATE

Export parity is evidence, not promotion

The selected ONNX graph passed the recorded PyTorch-versus-ONNX Runtime tolerance and 50-run representative CPU timing. Those completed facts replace a future-parity placeholder, but the artifact remains shadow-only and creates no order signals.

Provider abstains on at least

- model, schema, feature order, coordinate, row size, or tick-size mismatch
- insufficient, discontinuous, stale, incomplete, or rejected-quality history
- more than 1,024 rows, non-finite tensors, or runtime timeout/cancellation/crash
- unordered quantiles or low/high/POC/volume/delta invariant failure
- artifact-hash mismatch or unqualified physical-delta reconstruction

Monitoring and governance

- retain source, data, split, result, checkpoint, ONNX, metadata, and report hashes
- monitor input saturation, row counts, feed quality, abstention, latency, and calibration drift
- prohibit online weight updates, runtime model merging, and silent coordinate fallback
- keep a reversible version switch and null-provider fallback
- label forecasts as predictions and expose cutoff, horizon, age, model, and unavailable reason
- preserve the invariant that no live order-execution path is introduced

LITERATURE BOUNDARY

Related work - transferable mechanisms

A cited component motivates an experiment; it does not establish FDT-N performance.

TRANSFER

Time-series forecasting

TFT motivates direct quantiles, gates, and known-versus-observed covariates [S07]. PatchTST treats regular history as patches [S08]; here one sealed bar is already a domain patch, while strict channel independence is rejected. iTransformer, Crossformer, TimeXer, and Moirai motivate structured comparators, not native ragged-row support [S09-S12].

TRANSFER

Simple and normalized controls

Strong linear baselines can outperform complex forecasters [S05], so persistence, ridge, seasonal, and quantile-linear controls are mandatory. Reversible instance normalization is an ablation only; the implemented coordinate uses causal physical scales [S06].

TRANSFER

Limit-order-book models

DeepLOB, TransLOB, and TLOB motivate structural processing before temporal processing [S13-S15]. Their usual displayed-book and future-midprice tasks differ from executed-flow footprint marginals. FI-2010 and deep order-flow imbalance inform evaluation and baselines, not a footprint benchmark or P&L claim [S27-S28].

TRANSFER

Transformer components

Scaled attention, RMSNorm, SwiGLU, and relative-position work provide mature component precedents [S01-S04]. Qwen component choices are large-language-model precedent only [S20-S21]; language reasoning, grouped-query cache optimization, and agentic behavior do not transfer to a fixed supervised tensor map.

LITERATURE BOUNDARY

Related work - rejected scale mechanisms and calibration

The short exact sequence makes giant-model machinery disproportionate to the identified problem.

Mechanism	Original problem	FDT-N condition	Decision
MoE / expert routing	giant capacity specialization	no demonstrated bottleneck	reject v1
Latent / grouped-query attention	autoregressive KV-cache width	no generated token cache	reject
Native sparse attention	very long contexts	64 bars / 4,096 pairs	reject
Kimi linear recurrence	long-memory efficiency	reset and gap semantics dominate	defer
Runtime singular-value scaling	task adaptation	breaks frozen hash lineage	reject
Evolutionary model merging	combine qualified experts	no qualified expert pool	defer offline
LLM reasoning / agents	language generation and tools	fixed supervised map	out of scope

PROBABILISTIC CALIBRATION

Conformal correction is optional, not universal

Conformalized quantile regression can correct intervals on a held-out calibration set under its assumptions [S26]. Markets are serially dependent and nonstationary. Any later chronological conformal study must report empirical coverage under that exact procedure, not a universal guarantee. This frozen split has no separate calibration partition.

CONCLUSION

Claims held, supported conclusion, and next experiment

A complete artifact can narrow uncertainty without authorizing deployment or capital.

The paper does not claim

- full rows are better than aggregates until the registered ablation is run
- learned queries are semantically interpretable without intervention evidence
- the seven marginals form a coherent joint candle or physical-delta distribution
- results generalize across symbols, venues, intervals, sessions, or regimes
- forecast accuracy implies a directional, tradable, profitable, or risk-adjusted edge
- artifact parity implies production latency, operational safety, or capital suitability
- the model is state of the art, production-ready, or robust to distribution shift

SUPPORTED MECHANICAL CONCLUSION

What FDT-N is

FDT-N is a compact hierarchical direct multi-horizon transformer. It maps every accepted footprint row plus deterministic bar aggregates into ordered marginal quantiles for seven future auction targets, without autoregressive forecast feedback. The selected exported artifact is hash-bound, parity-checked, and shadow-only.

EMPIRICAL SCOPE

What this frozen study can say

Validation-selected seed 17 had pooled normalized q50-forecast MAE 0.23587: 7.25% below persistence (0.25430) but 2.68% above train-only ridge (0.22972). It beat persistence in 27/56 cells by point estimate and 21/56 by descriptive interval; corresponding ridge counts were 1/56 and 0/56. Central 80% coverage was 0.7898. Thus FDT-N did not beat the stronger linear control.

NEXT EVIDENCE

What remains before promotion

Run task-matched distributional baselines and registered ablations; preserve a later untouched outer period; add cross-symbol, cross-venue, and regime tests; validate live receive-time provenance and end-to-end latency; then conduct a separate execution study with frozen costs, risk, and abstention policy. Until then: no deployment and no capital allocation.

LINEAGE

Reproducibility ledger (1/2)

Full SHA-256 identities bind the source narrative, data, splits, baselines, and every seed.

Artifact	Seed	Bytes	SHA-256
Dataset manifest	shared	4,790	b88d89565febb97dbe56151fe447e3952fa844b8c433ecb86cb747e11e4ce412
Encoded dataset	shared	9,963,548	d83f78894af08b65123d2bf9f4fbef8ca28e901a552d3af2c61f6e37513bf6cc
Research paper	shared	57,257	8f56a7dc1320068b1c566c16ba82915e4bd5658356c265082365c40945cf22f0
Evidence ledger	shared	33,309	f67a45e909c72f12814f5e9ed8014ed59c5cc097729d075e625e6bb0355cc0f0
Selected model metadata	17	3,615	9f998ac6b56b008ac02aaa0ed5c4f08447b1735f33027294773d9e1aa0867763
Selected ONNX model	17	1,834,205	adce8cd39312bf3ad1389a23ed5ca24d6f47a7f58dbb26d76ae7d54af1011af7
Results JSON	17	119,448	746403a05790753943b8c22f615f097c8dc397e413be49f35ea553b82e0cbd0a
Checkpoint	17	1,519,582	9d2cf525971f81fc93ca9ed97103cffb29203d7fd59edea5bd92d0e1480e124a
Split indices	17	30,658	446aa8541734c39956a531e24cbd23dec0e3ca6720b5845af1e3d440d4078d7c
Direct ridge	17	31,552	49f4e92d0dba5dacf804b08eac4c9011b174e2e6c313df4ed7d6e948676a38aa
Results JSON	29	117,655	1b3e19b476b65e27becb30aaf600ea0384477b64b13f4d0d3eb2abddf3d0e0cc
Checkpoint	29	1,519,582	1800197b49ae6ccb58b6e7cf7eb4c9d90708f87e95233a16d1f32098b3825036
Split indices	29	30,658	446aa8541734c39956a531e24cbd23dec0e3ca6720b5845af1e3d440d4078d7c

Primary selection rule: minimum validation loss, tie-broken by seed number; no test score enters selection. Results record seeds, device class, configuration, elapsed time, and one post-selection test evaluation per seed. Exact GPU model and original command lines were not frozen, and the repository had unrelated concurrent dirty work; there is no separate FDT-N outer holdout. These are disclosed reproducibility limits.

LINEAGE

Reproducibility ledger (2/2)

Full SHA-256 identities bind the source narrative, data, splits, baselines, and every seed.

Artifact	Seed	Bytes	SHA-256
Direct ridge	29	31,552	49f4e92d0dba5dacf804b08eac4c9011b174e2e6c313df4ed7d6e948676a38aa
Results JSON	43	119,272	4d28b1e41c6f93e23f6c86261d7710a9974a58c5985aab7c029415f2a4e605ad
Checkpoint	43	1,519,582	439298be69f270bb7222733d1c483cdd6272c7700b02fb0be11f031012df8ed6
Split indices	43	30,658	446aa8541734c39956a531e24cbd23dec0e3ca6720b5845af1e3d440d4078d7c
Direct ridge	43	31,552	49f4e92d0dba5dacf804b08eac4c9011b174e2e6c313df4ed7d6e948676a38aa

Primary selection rule: minimum validation loss, tie-broken by seed number; no test score enters selection. Results record seeds, device class, configuration, elapsed time, and one post-selection test evaluation per seed. Exact GPU model and original command lines were not frozen, and the repository had unrelated concurrent dirty work; there is no separate FDT-N outer holdout. These are disclosed reproducibility limits.

LIMITATIONS

Decision boundary and required next evidence

A complete PDF is a reporting artifact, not model promotion or authorization.

What this report establishes

- Exact identities for the frozen data and every run
- Validation-only primary selection across declared seeds
- Per-cell marginal scores and point-baseline effects
- Bound clipping audit and selected ONNX parity evidence

What it does not establish

- A joint candle or physical-delta distribution
- Cross-venue, cross-symbol, or cross-regime generality
- Economic value after fills, latency, fees, and impact
- Production latency, operational safety, or capital suitability

Required before promotion

- Independent later outer holdout and calibration period
- Task-matched distributional baselines and ablations
- Live provider latency, memory, drift, and failure tests
- Separate execution study with frozen risk and cost policy

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